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SSH Log Analysis

Splunk Basics – SSH Log Analysis

STEP-BY-STEP LAB REPORT

OBJECTIVE

The objective of this lab is to learn how **SSH authentication logs can be analyzed using Splunk** to detect suspicious login activity.

In this lab we will:

- Upload **Zeek-style SSH logs** into Splunk.
- Analyze SSH authentication attempts.
- Detect **failed login attempts and successful connections**.
- Identify **possible brute force attacks or unauthorized access**.

SSH log monitoring is important because attackers often attempt to **gain access to servers using repeated login attempts**.

TOOLS AND ENVIRONMENT

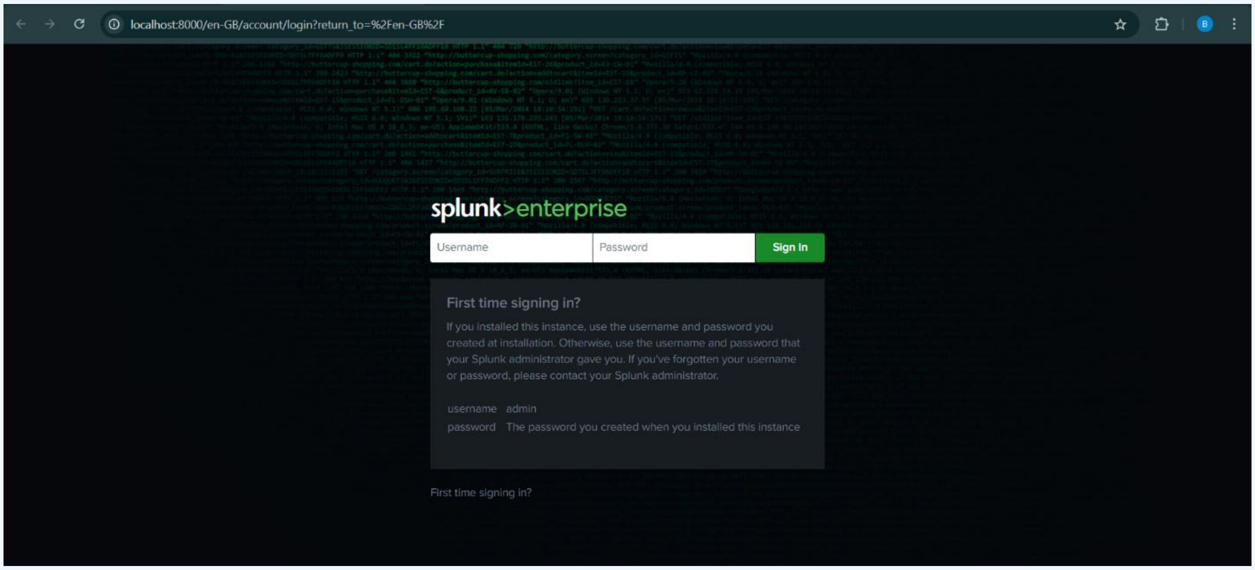
Component	Description
Splunk	Log management and data analysis platform
Log Source	Zeek SSH logs
File Format	JSON
Index Used	ssh_lab
Sourcetype	json

LAB SETUP

Before performing the analysis, the SSH log file must be uploaded to Splunk.

Step 1 – Open Splunk Web

1. Open **Splunk Web Interface** in your browser.
2. Log in using your credentials.

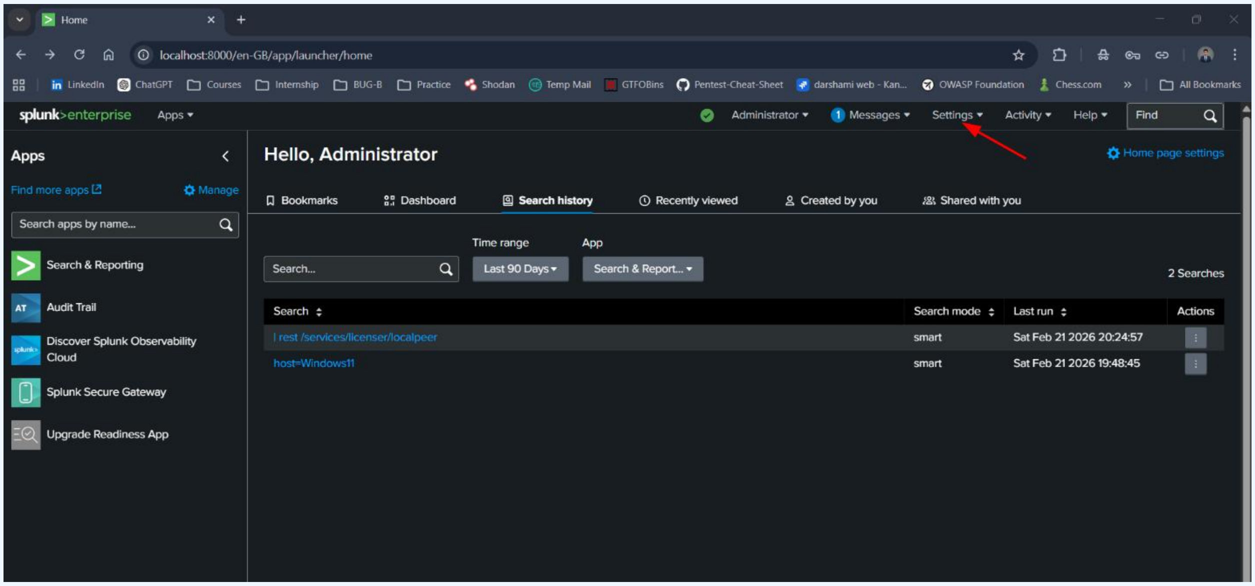


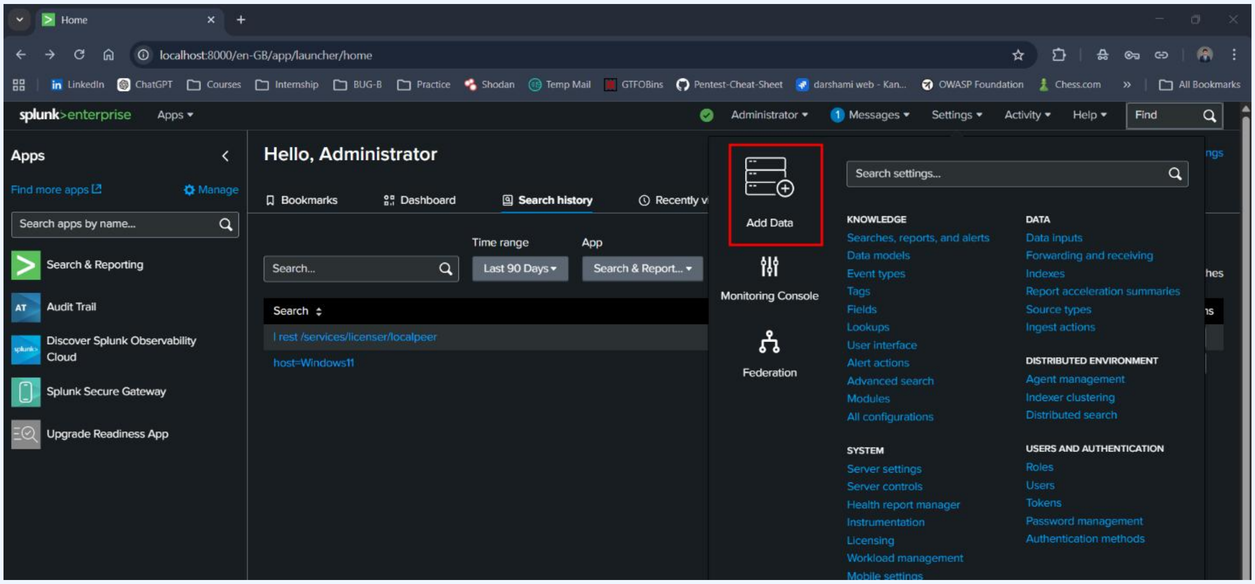
Step 2 – Navigate to Add Data

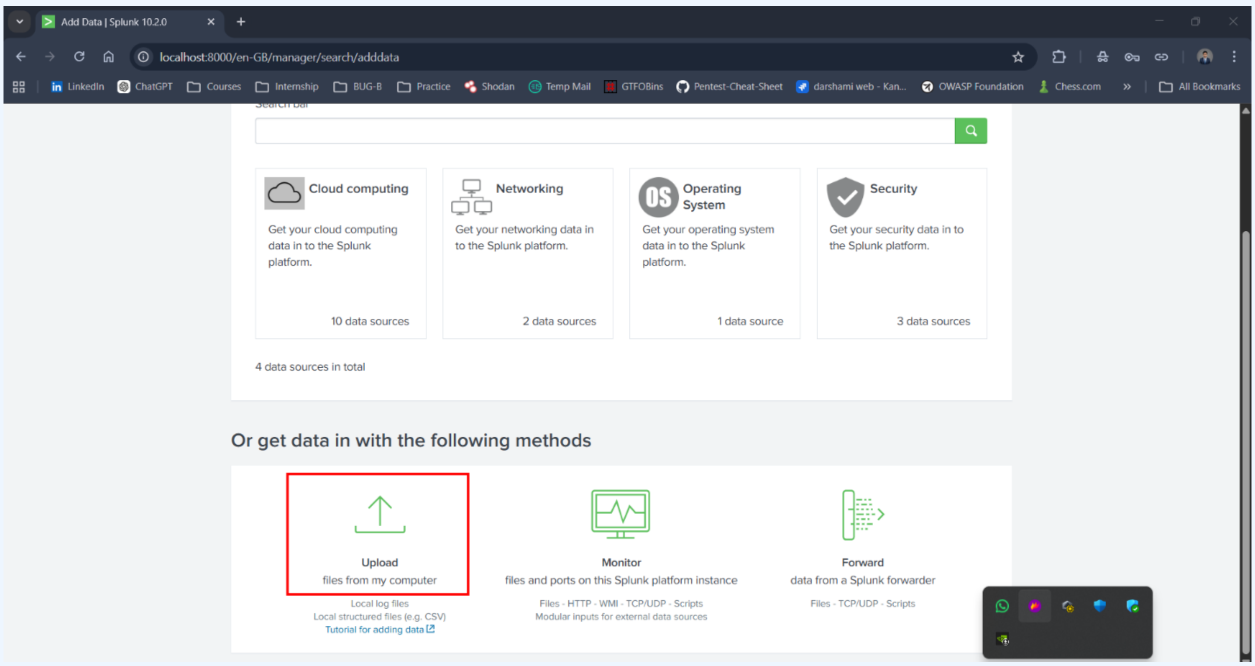
1. Go to:

Settings → Add Data

2. Select **Upload**.





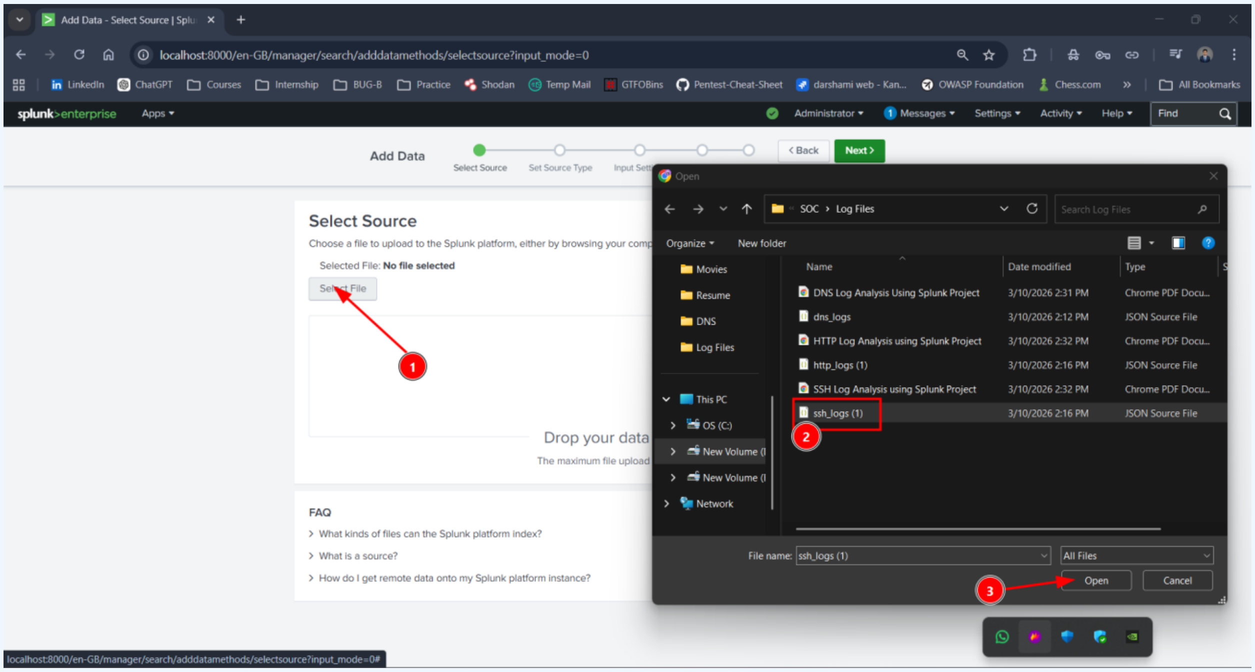


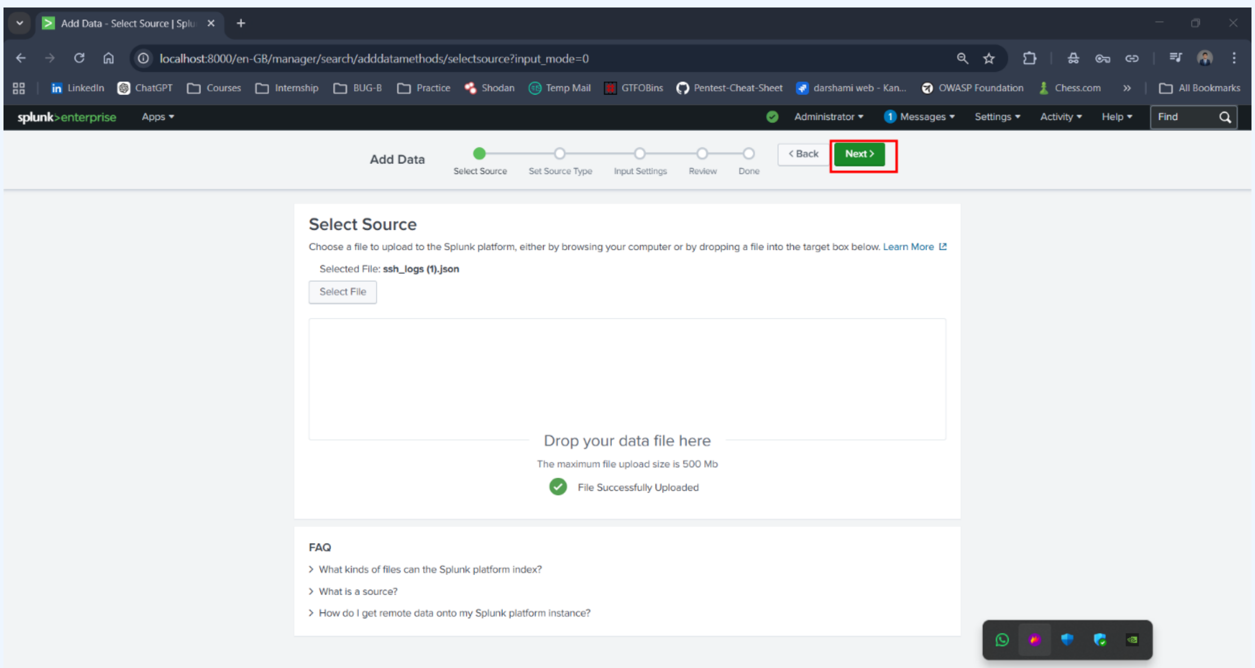
Step 3 – Upload the SSH Log File

1. Click **Select File**
2. Upload the file:

```
ssh_logs
```

This file contains **SSH connection events captured in Zeek-style format.**


Add Data - Select Source | Splu
localhost:8000/en-GB/manager/search/adddatamethods/selectsource?input_mode=0
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
< Back
Next >
Select Source
Selected File: No file selected
Select File
1
Drop your data
Open
SOC > Log Files
Search Log Files
Organize
New folder
Movies
Resume
DNS
Log Files
This PC
OS (C:)
Network
Name
Date modified
Type
DNS Log Analysis Using Splunk Project
3/10/2026 2:31 PM
Chrome PDF Docu...
dns_logs
3/10/2026 2:12 PM
JSON Source File
HTTP Log Analysis using Splunk Project
3/10/2026 2:32 PM
http_logs (1)
3/10/2026 2:16 PM
SSH Log Analysis using Splunk Project
ssh_logs (1)
2
File name:
All Files
3
Open
Cancel
FAQ
What kinds of files can the Splunk platform index?
What is a source?
How do I get remote data onto my Splunk platform instance?


Add Data - Select Source | Splu
localhost:8000/en-GB/manager/search/adddatamethods/selectsource?input_mode=0
splunk>enterprise
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
< Back
Next >
Select Source
Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. Learn More
Selected File: ssh_logs (1).json
Select File
Drop your data file here
The maximum file upload size is 500 Mb
File Successfully Uploaded
FAQ
What kinds of files can the Splunk platform index?
What is a source?
How do I get remote data onto my Splunk platform instance?

Step 4 – Configure Source Type

Set the source type as:

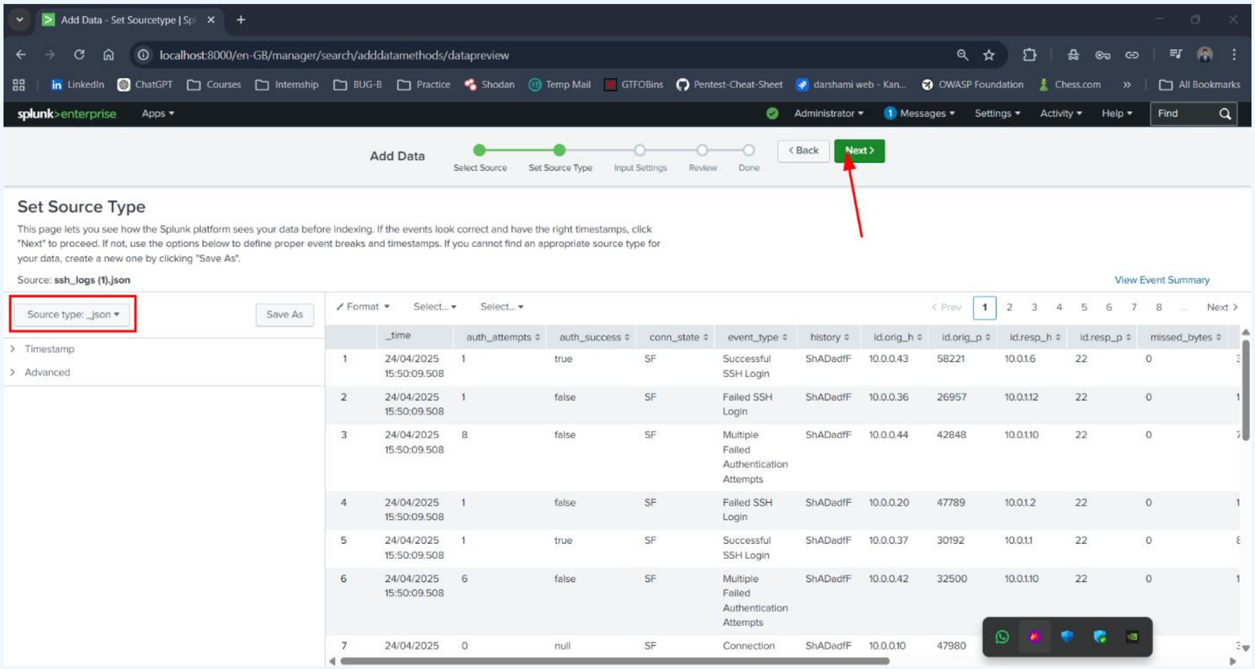
```
_json
```

or create a custom sourcetype:

```
ssh_logs
```

Reason:

- The log file is structured in **JSON format**.
- Splunk automatically extracts fields such as:
  - id.orig_h
  - auth_success
  - event_type
  - timestamp



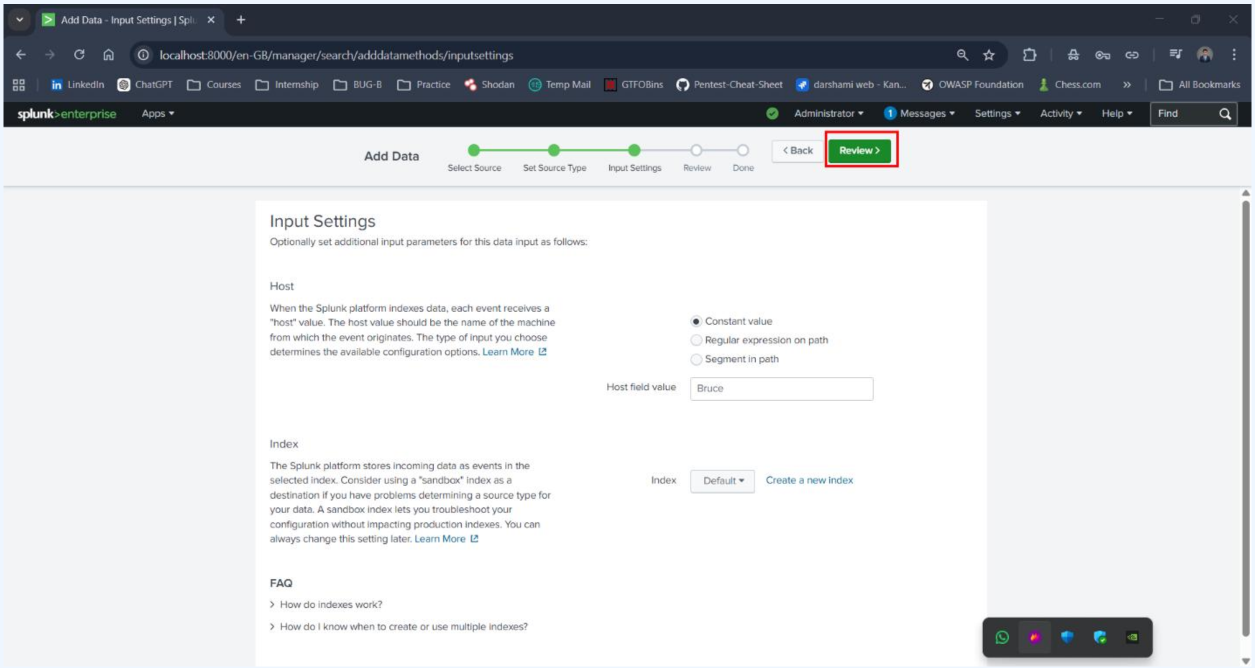
Step 5 – Configure Index

Create or select an index:

```
ssh_lab
```

Reason:

- Keeps SSH lab data separated from other datasets.
- Makes analysis easier.



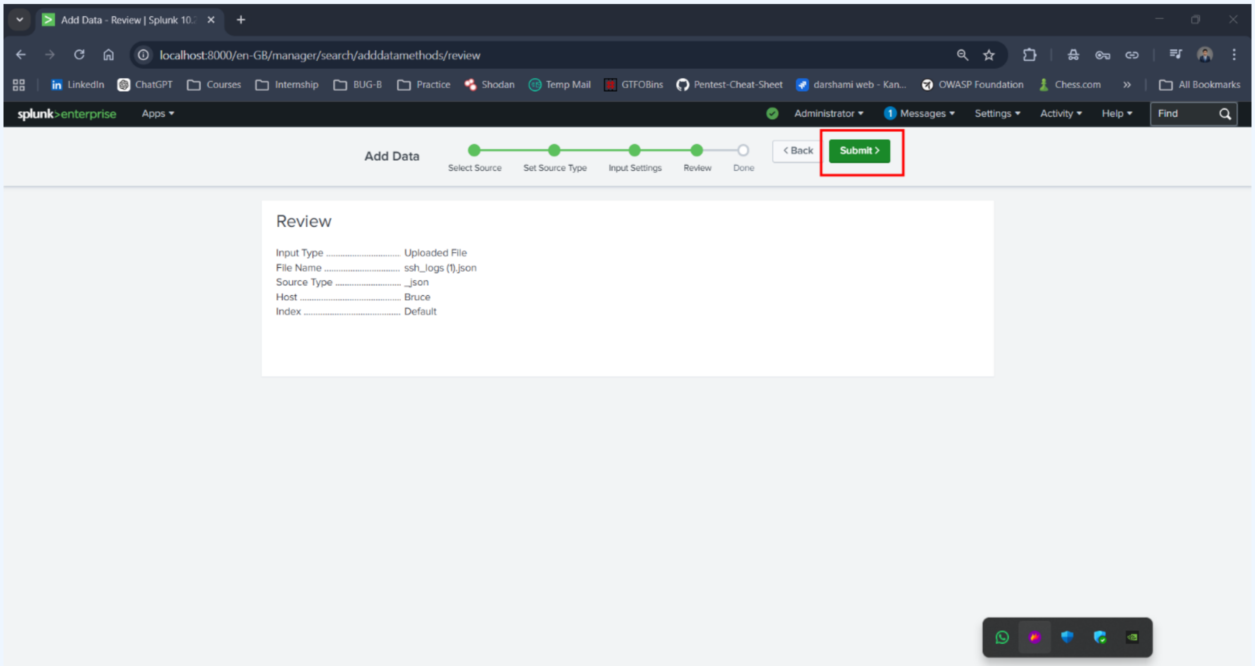
Step 6 – Complete Upload

1. Review settings.
2. Click **Submit**.
3. Splunk will ingest and index the data.

To confirm indexing run:

```
index=ssh_lab
```

If events appear, the data upload was successful.



UNDERSTANDING SSH LOGS

SSH logs record **authentication attempts and connections to servers**.

Common fields in Zeek SSH logs:

Field	Description
id.orig_h	Source IP attempting SSH connection
id.resp_h	Destination server IP
auth_success	Indicates whether authentication succeeded
event_type	Type of SSH event
timestamp	Time of the event

Example SSH Log Event

```
{
"id.orig_h":"192.168.1.10",
"id.resp_h":"192.168.1.5",
"auth_success":false,
"event_type":"failed"
}
```

LAB TASKS

Task 1: List Top 10 Endpoints with Failed SSH Login Attempts

SPL Query

```
index=ssh_lab sourcetype="json" auth_success=false
| stats count by "id.orig_h"
| sort -count
| head 10
```

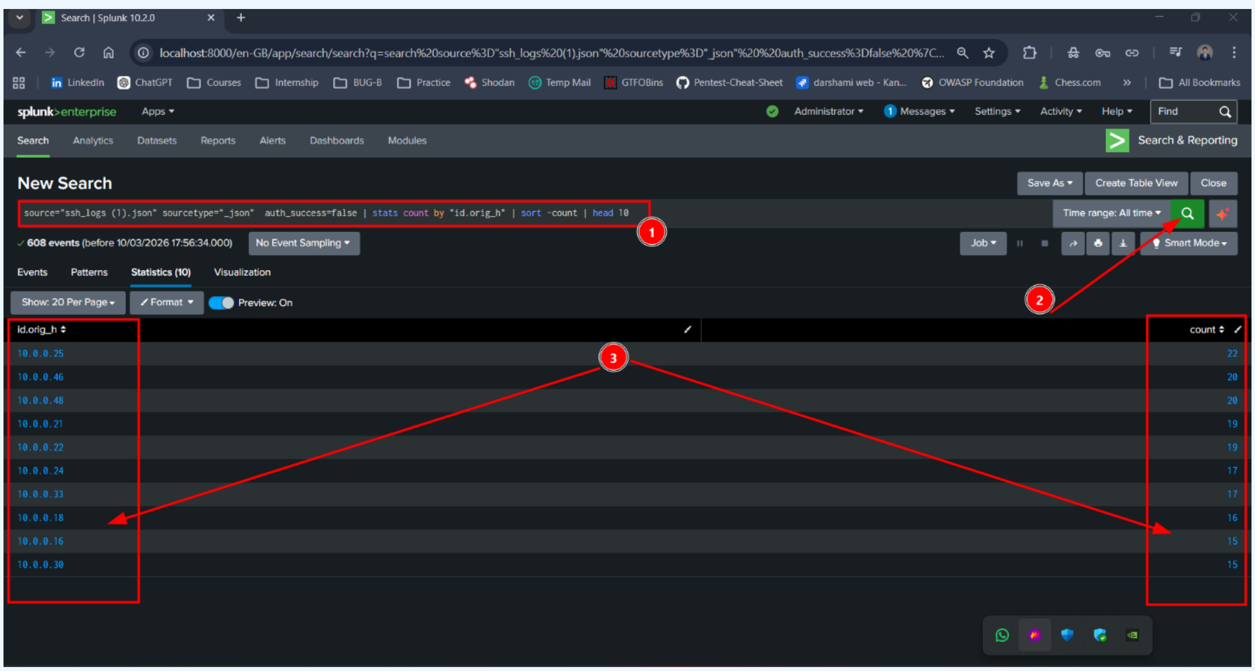
Explanation

Command	Purpose
index=ssh_lab	Searches logs in ssh_lab index
sourcetype="json"	Filters JSON formatted logs
auth_success=false	Shows only failed login attempts
stats count by id.orig_h	Counts failed attempts by source IP
sort -count	Sorts results in descending order
head 10	Displays top 10 IP addresses

Security Insight

This query helps detect:

- **Brute force attacks**
- IPs attempting repeated SSH logins
- Suspicious external access attempts



Task 2: Find the Total Number of SSH Connections

SPL Query

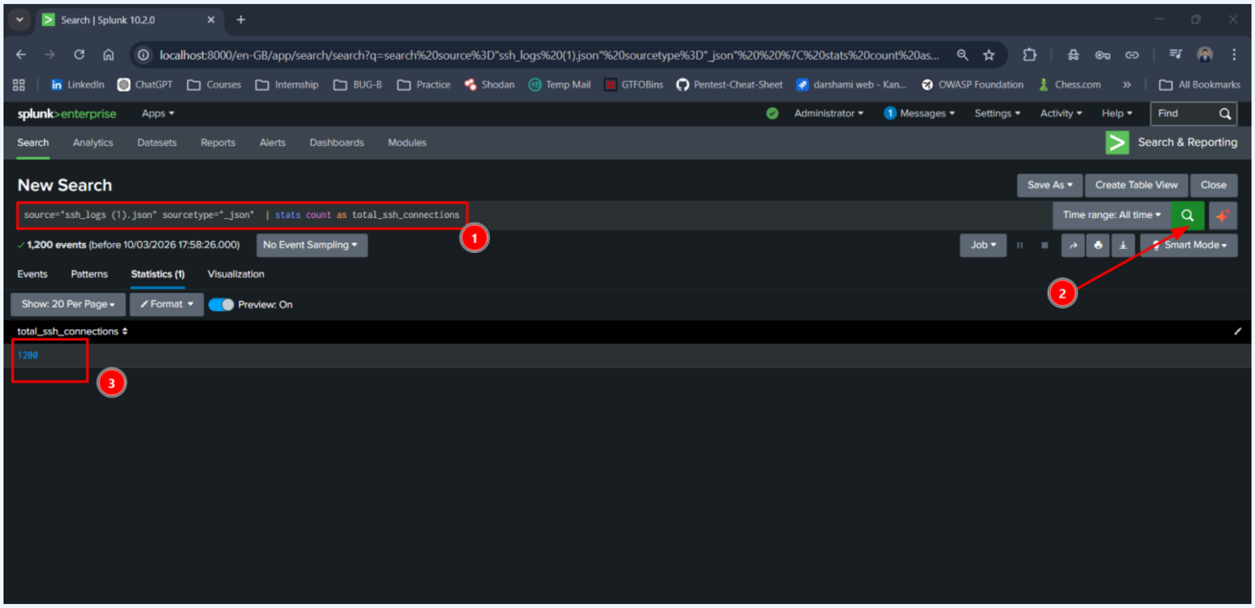
```
index=ssh_lab sourcetype="json"
| stats count as total_ssh_connections
```

Explanation

Command	Purpose
stats count	Counts all events
as total_ssh_connections	Renames the output field

What This Shows

This query provides the **total number of SSH connection events** recorded in the logs.



This helps understand **overall SSH activity** on the network.

Task 3: Count All Event Types in the Logs

SPL Query

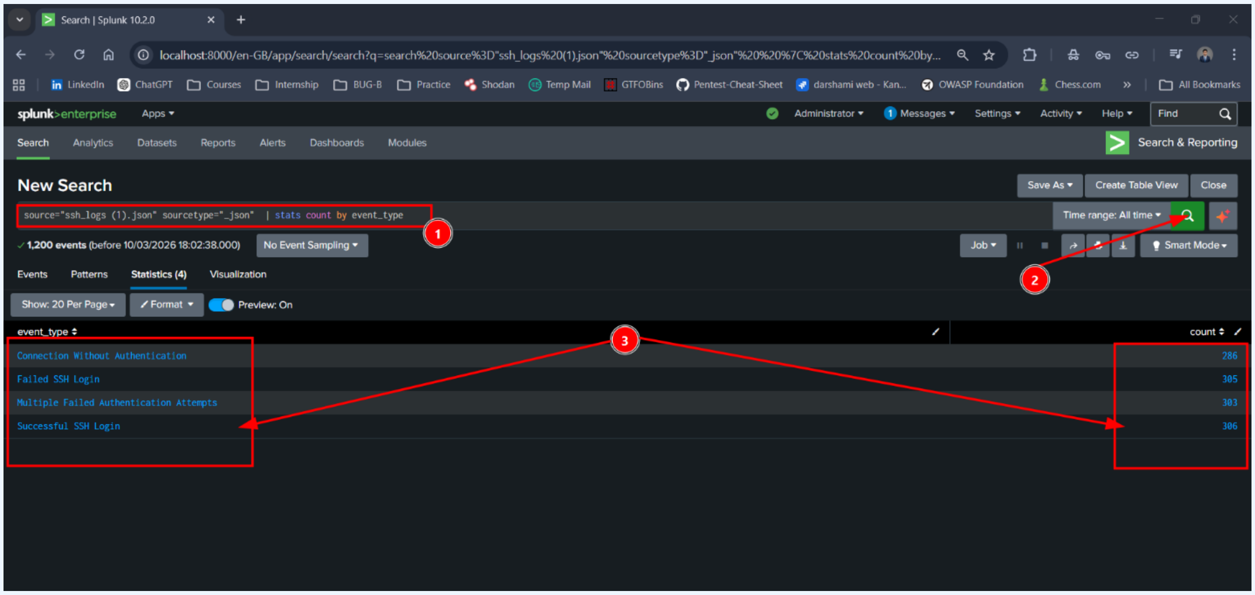
```
index=ssh_lab sourcetype="json"
| stats count by event_type
```

Explanation

Command	Purpose
stats count by event_type	Groups events based on event type

Possible Event Types

Event Type	Meaning
successful	Successful SSH authentication
failed	Failed login attempt
no-auth	Connection attempt without authentication
multiple-failed	Multiple failed attempts from same source



SECURITY INSIGHTS

Analyzing SSH logs can help detect:

1. **Brute Force Attacks:** Large numbers of **failed login attempts from one IP**.
2. **Unauthorized Access:** Unexpected **successful logins from unknown IP addresses**.
3. **Suspicious Login Patterns**

Examples:

- Many failed attempts followed by one successful login
- Logins at unusual times
- Connections from unfamiliar geographic locations

Splunk enables **real-time monitoring and threat detection** through log analysis.

CONCLUSION

In this lab we successfully:

- Uploaded **Zeek SSH logs into Splunk**
- Learned how Splunk processes **JSON log data**
- Used **SPL queries** to analyze SSH authentication activity

The analysis helped identify:

- Top IP addresses generating failed login attempts
- Total number of SSH connections
- Distribution of SSH event types

This type of log analysis is critical for **detecting brute force attacks, monitoring server access, and improving system security**.